

## SAFETY DATA SHEET

Creation Date 09-January-2012

Revision Date 27-March-2024

Revision Number 3

### 1. Identification

**Product Name** Cerium(IV) oxide, REacton®

**Cat No. :** 40581

**CAS-No** 1306-38-3  
**Synonyms** Cerium dioxide

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Not classified under the Hazardous Products Regulations (SOR/2015-17)

Based on available data, the classification criteria are not met

#### Label Elements

None required

### 3. Composition/Information on Ingredients

| Component    | CAS-No    | Weight % |
|--------------|-----------|----------|
| Cerium oxide | 1306-38-3 | >95      |

#### 4. First-aid measures

|                                                               |                                                                                                                                          |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                                            | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.                                                 |
| <b>Skin Contact</b>                                           | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes.                                    |
| <b>Inhalation</b>                                             | Remove from exposure, lie down. Remove to fresh air.                                                                                     |
| <b>Ingestion</b>                                              | Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Drink plenty of water. If possible drink milk afterwards. |
| <b>Most important symptoms/effects<br/>Notes to Physician</b> | No information available.<br>Treat symptomatically                                                                                       |

#### 5. Fire-fighting measures

|                                         |                                                                                                         |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                |
| <b>Flash Point</b>                      | No information available                                                                                |
| <b>Method -</b>                         | No information available                                                                                |
| <b>Autoignition Temperature</b>         | No information available                                                                                |
| <b>Explosion Limits</b>                 |                                                                                                         |
| <b>Upper</b>                            | No data available                                                                                       |
| <b>Lower</b>                            | No data available                                                                                       |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                |

#### Specific Hazards Arising from the Chemical

Keep product and empty container away from heat and sources of ignition.

#### Hazardous Combustion Products

Thermal decomposition can lead to release of irritating gases and vapors.

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

#### NFPA

**Health**  
0

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

#### 6. Accidental release measures

|                                  |                                                                             |
|----------------------------------|-----------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Ensure adequate ventilation. Use personal protective equipment as required. |
| <b>Environmental Precautions</b> | See Section 12 for additional Ecological Information.                       |

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal.

#### 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                     |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Avoid contact with skin and eyes. Avoid contact with skin and clothing. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Avoid breathing vapors or mists. Do not ingest. If swallowed then seek immediate medical assistance. Wash thoroughly after handling. |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Storage.** Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Incompatible Materials. Strong oxidizing agents. Oxidizing agent.

## 8. Exposure controls / personal protection

**Exposure Guidelines** This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.

**Engineering Measures** None under normal use conditions.

### Personal protective equipment

**Eye Protection** Wear safety glasses with side shields (or goggles)  
**Hand Protection** Protective gloves

| Glove material                                      | Breakthrough time                 | Glove thickness | Glove comments         |
|-----------------------------------------------------|-----------------------------------|-----------------|------------------------|
| Natural rubber<br>Nitrile rubber<br>Neoprene<br>PVC | See manufacturers recommendations | -               | Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### **Respiratory Protection**

No protective equipment is needed under normal use conditions.

**Recommended Filter type:** Particle filter

### Environmental exposure controls

No information available.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Physical State</b>                   | Powder Solid             |
| <b>Appearance</b>                       | Light yellow             |
| <b>Odor</b>                             | Odorless                 |
| <b>Odor Threshold</b>                   | No information available |
| <b>pH</b>                               | No information available |
| <b>Melting Point/Range</b>              | 2600 °C / 4712 °F        |
| <b>Boiling Point/Range</b>              | No information available |
| <b>Flash Point</b>                      | No information available |
| <b>Evaporation Rate</b>                 | Not applicable           |
| <b>Flammability (solid,gas)</b>         | No information available |
| <b>Flammability or explosive limits</b> |                          |
| <b>Upper</b>                            | No data available        |
| <b>Lower</b>                            | No data available        |
| <b>Vapor Pressure</b>                   | No information available |

|                                        |                          |
|----------------------------------------|--------------------------|
| Vapor Density                          | Not applicable           |
| Specific Gravity                       | 7.130                    |
| Solubility                             | Insoluble in water       |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | Not applicable           |
| Molecular Formula                      | Ce O <sub>2</sub>        |
| Molecular Weight                       | 172.12                   |

## 10. Stability and reactivity

|                                  |                                                                          |
|----------------------------------|--------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                               |
| Stability                        | Stable under normal conditions.                                          |
| Conditions to Avoid              | Incompatible products.                                                   |
| Incompatible Materials           | Strong oxidizing agents, Oxidizing agent                                 |
| Hazardous Decomposition Products | Thermal decomposition can lead to release of irritating gases and vapors |
| Hazardous Polymerization         | No information available.                                                |
| Hazardous Reactions              | None under normal processing.                                            |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

| Component    | LD50 Oral           | LD50 Dermal         | LC50 Inhalation        |
|--------------|---------------------|---------------------|------------------------|
| Cerium oxide | >5000 mg/kg ( Rat ) | >2000 mg/kg ( Rat ) | >5.05 mg/L ( Rat ) 4 h |

**Toxicologically Synergistic** No information available

#### Products

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component    | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------|-----------|------------|------------|------------|------------|------------|
| Cerium oxide | 1306-38-3 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** Not mutagenic in AMES Test

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains. .

| Component    | Freshwater Algae | Freshwater Fish                           | Microtox   | Water Flea         |
|--------------|------------------|-------------------------------------------|------------|--------------------|
| Cerium oxide | Not listed       | LC50 >100 mg/L/96h<br>(Brachydanio rerio) | Not listed | EC50 >100 mg/L/48h |

**Persistence and Degradability** Insoluble in water

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

**DOT** Not regulated  
**TDG** Not regulated  
**IATA** Not regulated  
**IMDG/IMO** Not regulated

## 15. Regulatory information

### International Inventories

| Component    | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|--------------|-----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Cerium oxide | 1306-38-3 | X   | -    | X    | ACTIVE                                        | 215-150-4 | -      | -   |

| Component    | CAS-No    | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|--------------|-----------|-------|----------|------|------|------|------|-------|-------|
| Cerium oxide | 1306-38-3 | X     | KE-05392 | X    | X    | X    | X    | X     | X     |

### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

### Other International Regulations

Authorisation/Restrictions according to EU REACH

Not applicable

Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component    | CAS-No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Cerium oxide | 1306-38-3 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component    | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Cerium oxide | 1306-38-3 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

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**Print Date**

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**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

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**End of SDS**